

JKBOSE 2026

CLASS 11TH

PLANT KINGDOM

LECTURE-07

BOTANY

ARSHID BHAT SIR



Topics To Covered

- 1** Gymnosperms ✓
- 2** Angiosperms (Introduction) ✓

GYMNOSPERMS

- ≡ Study of gymnosperms is kld "Gymnospermology"
- ≡ term Gymnosperm was Coined by Theoprastus (father of Botany).
- ≡ Gymnosperms include
 - ↙ Tall trees
 - ↘ medium Sized trees
 - ↘ Shrubs
- ≡ Tallest gymnosperm - Sequoia
(Giant red wood tree)
- ≡ Gymnosperm - Smallest plant group.

Habitat: Himalayan regions.



Main plant Body: Sporophyte ($2n$) — produces spores (n) by meiosis.



It consists of

-
- a) True root
 - b) True Stem
 - c) True leaves

- Vascular bundle (xylem & phloem) are $+$ nt.
- xylem lacks — vessels.
- phloem lacks — Companion cells.

Note: Ephedra
Welwetschia
Eyneturm

xylem possess true vessels.

≡ gymnosperm → gymno's = naked
→ Sperma = seed

• In gymnosperms, Seed remains Naked before and after fertilisation. Because of absence of ovary.

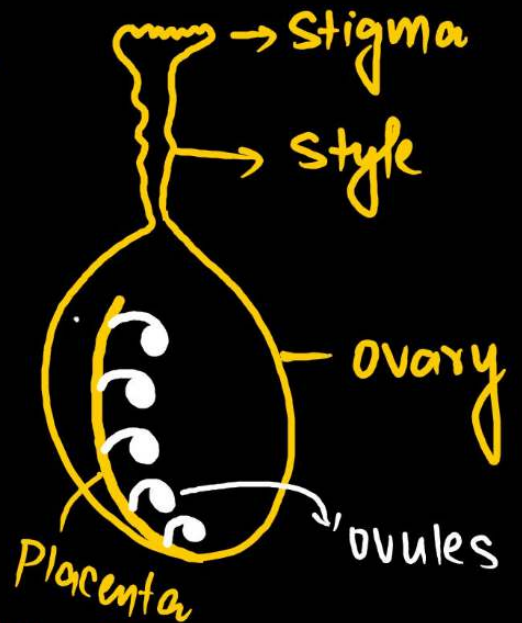
≡ In gymnosperms — ovule is +nt which forms seed after fertilisation.



"Angiosperms"

Note:-

Gymnosperms
↓
Pistil - nt



After fertilisation:-

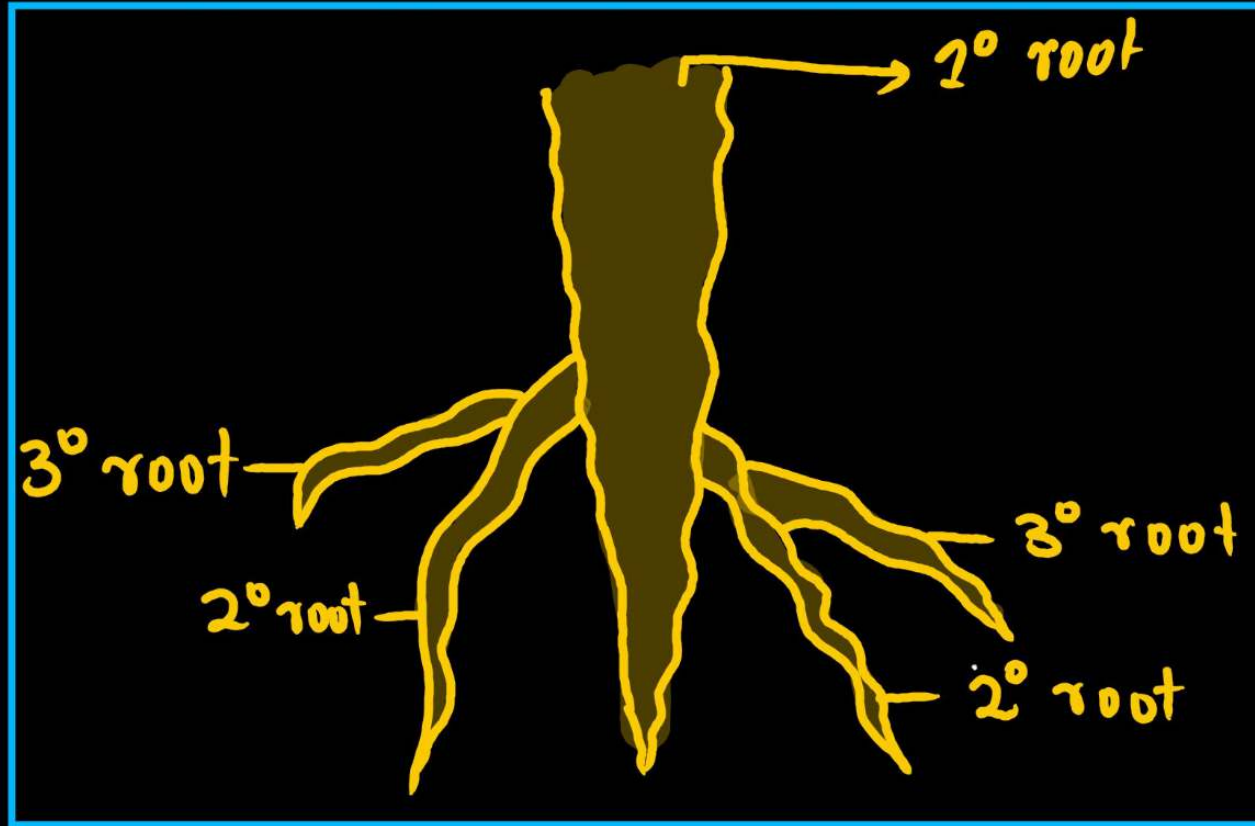
- ovary forms fruit
- ovule forms seed

Root : • Generally tap root.

1° root along with
lateral branches of
various orders (2°, 3°...)

Micorrhiza - Root associated with
fungi ex. Pinus.

Corraloid - Root associated with
N₂-fixing cyanobacteria
ex. Cycas.



Stem:

Branched
Ex. Pinus

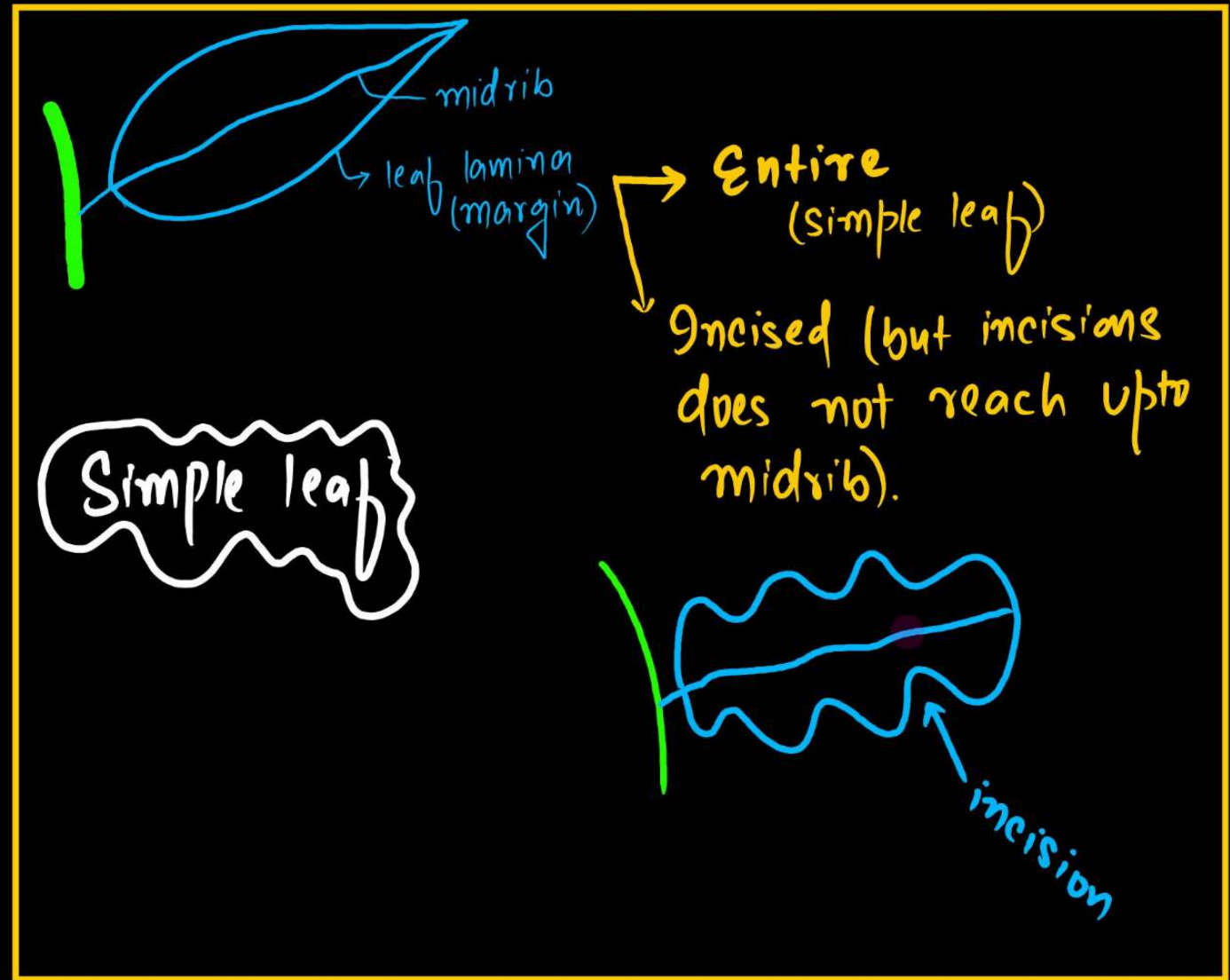
Unbranched
Ex. Cycas



Compound leaf

Incision upto midrib

leaflets

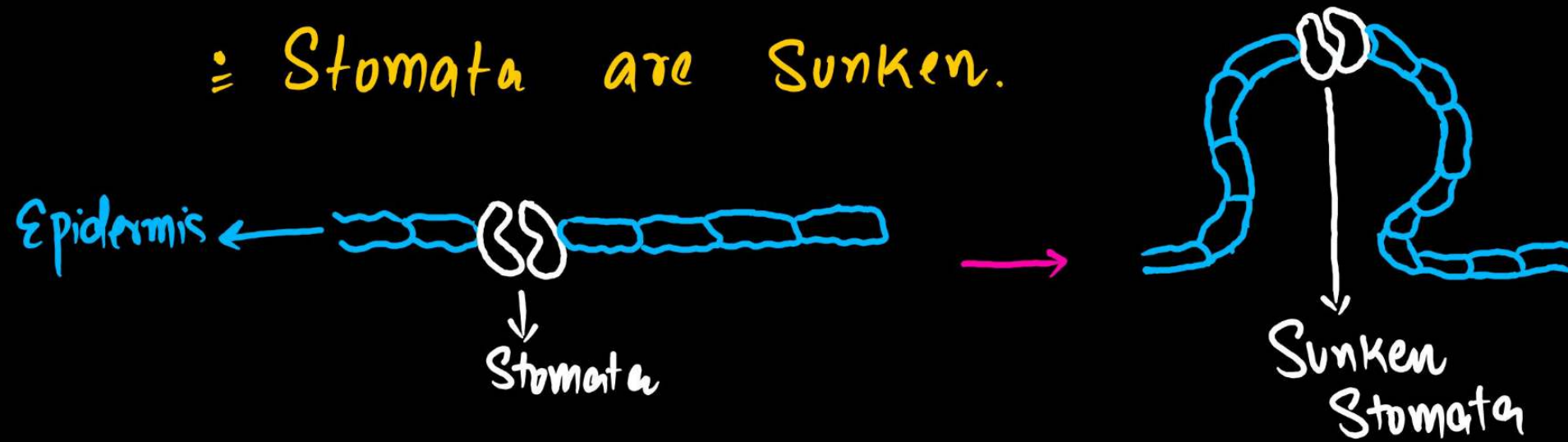


• In Gymnosperms - leaves can withstand high temp, humidity etc.

• Pinus - needle like leaves (persistent for years)

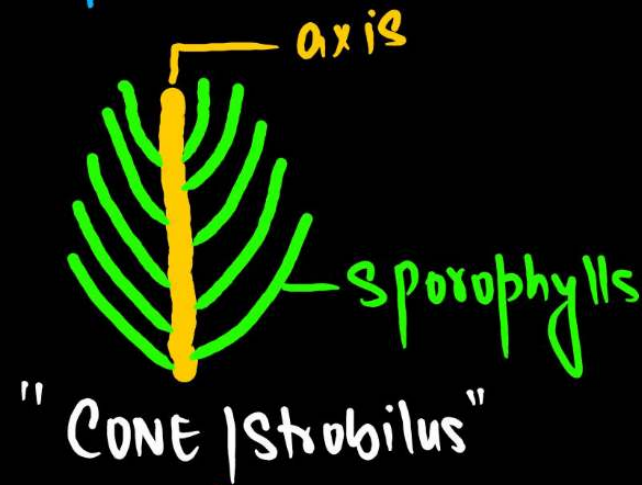
• leaves ^{possess} thick cuticle to avoid H_2O loss.

• Stomata are sunken.



• Gymnosperms are heterosporous $\rightarrow$ Microspore
 $\rightarrow$ Megaspore.

• Gymnosperms possess special leaves called as Sporophylls



Arranged compactly or loosely (i.e, lax) along an axis to form a Cone or Strobilus.

Two types:

- Male Cone | pollen cone | microsporangiate Cone.
- Female Cone | Seed cone | megasporangiate Cone.

A) MALE CONE:



↓
Possess Small sized Sporophylls
Called as microsporophylls.



↓
bears microsporangium

↓
possess microspore mother cell ($2n$)

↓ meiosis (inside microsporangium)

Microspores (n)

↓ develop inside microsporangium (onsitu development)

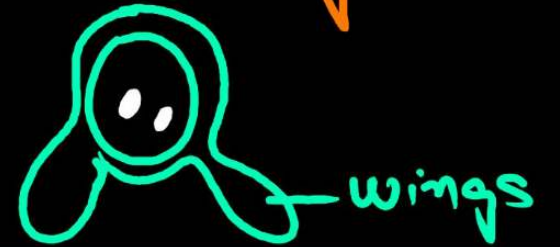
2 σ gametes



Pollen Grain (Reduced σ gametophyte)

↳ possess two σ gametes.

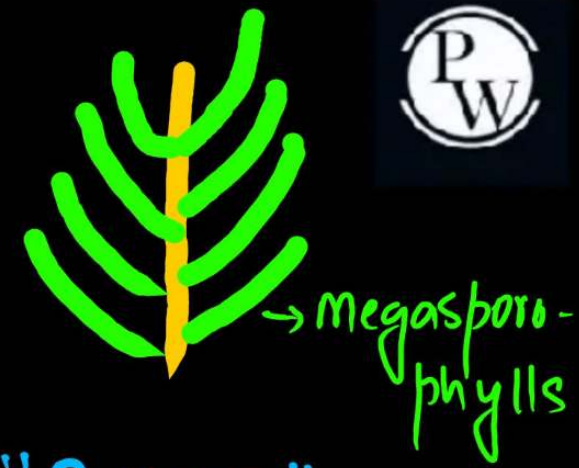
Pinus - possess winged pollen grain



B) FEMALE CONE:-

↓
Possess large sized special leaves

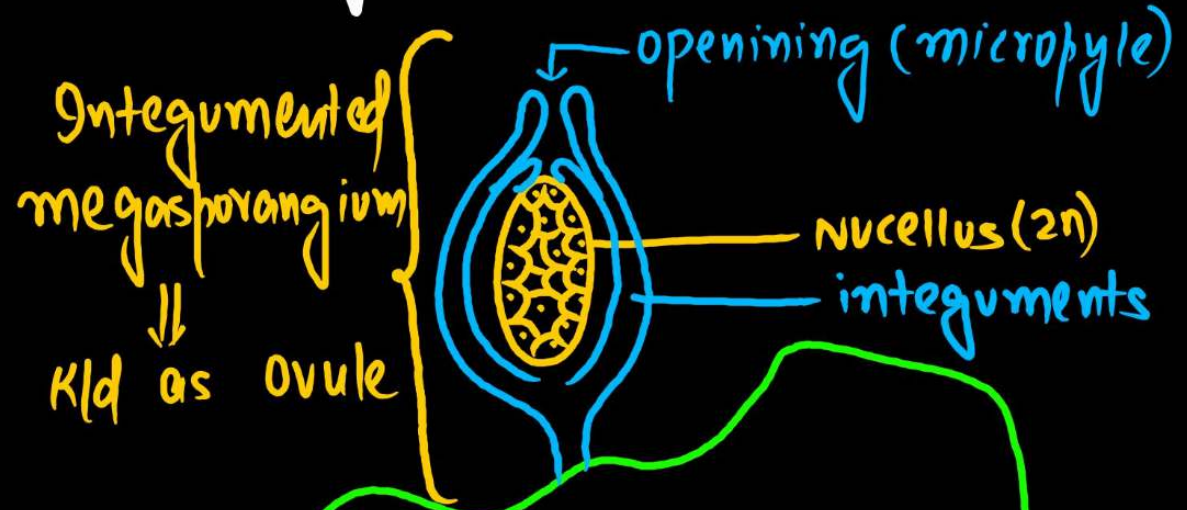
Called as megasporophylls



"♀ CONE"

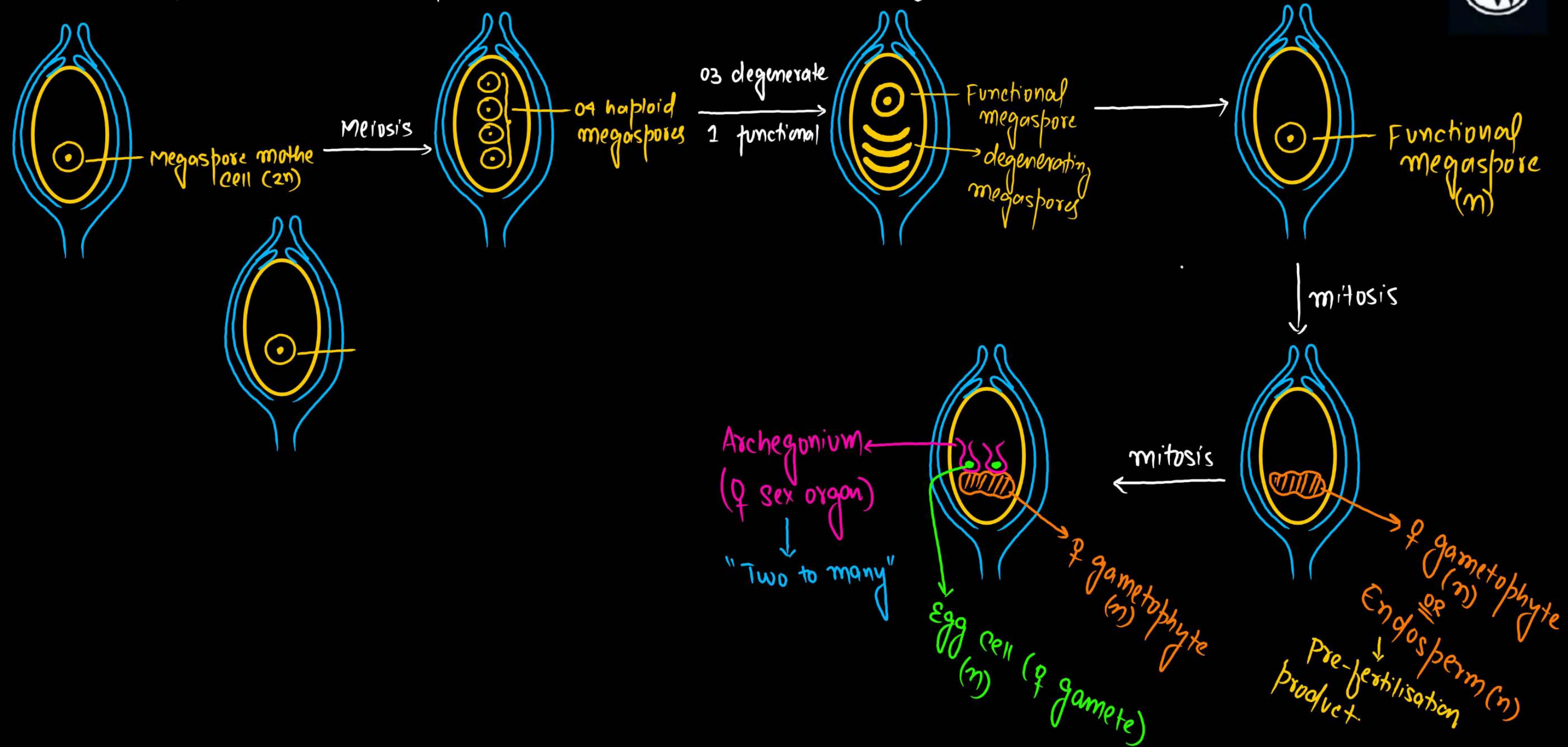
Possess megasporangium

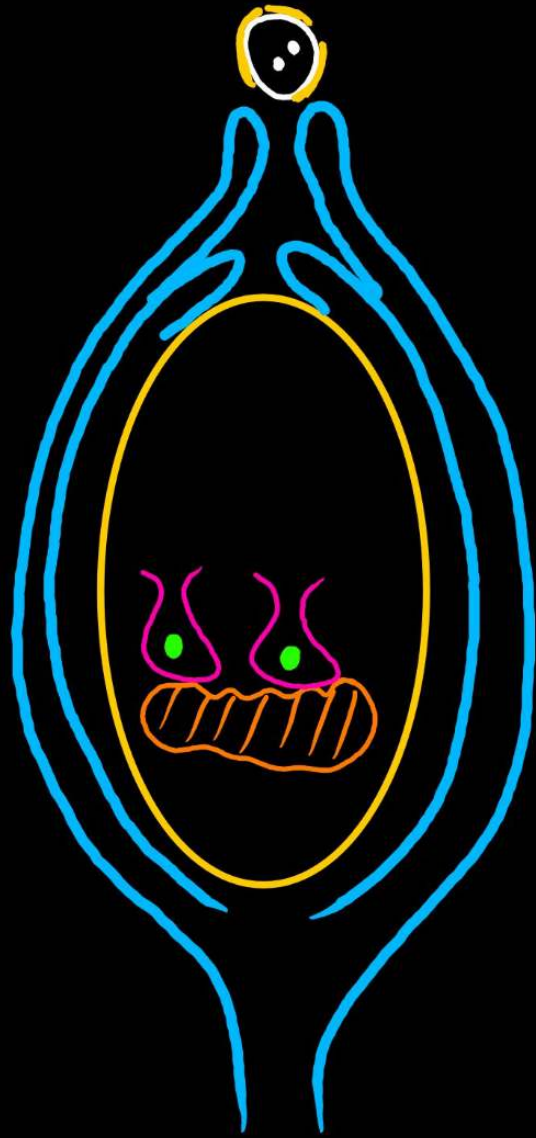
- It is ovule having
- a. Diploid tissue kld nucellus.
 - b. protective covering kld integuments



→ megasporophyll

≡ one of the cells of Nucellus — behaves as megaspore mother cell ($2n$)



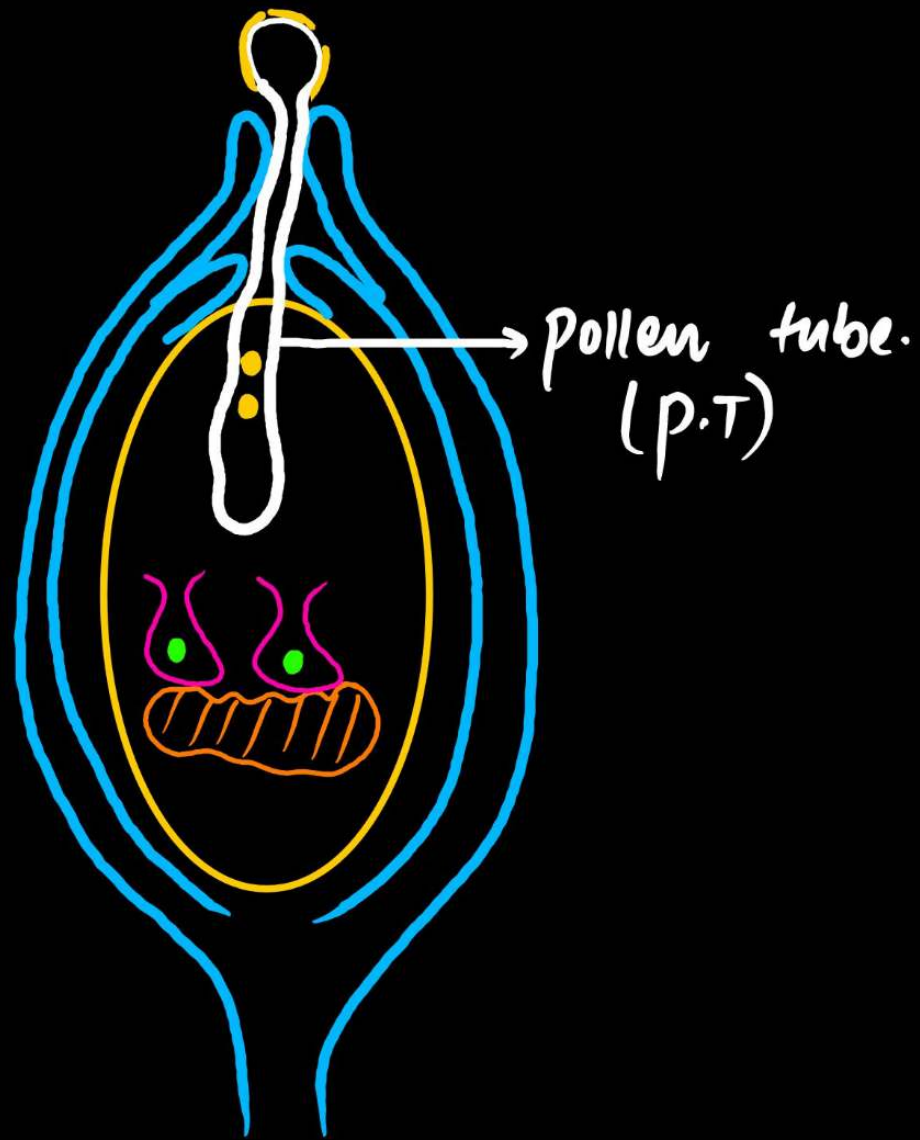


Pollination :-

→ Transfer of Pollen grains from microsporangium to opening of Ovule (i.e., micropyle)

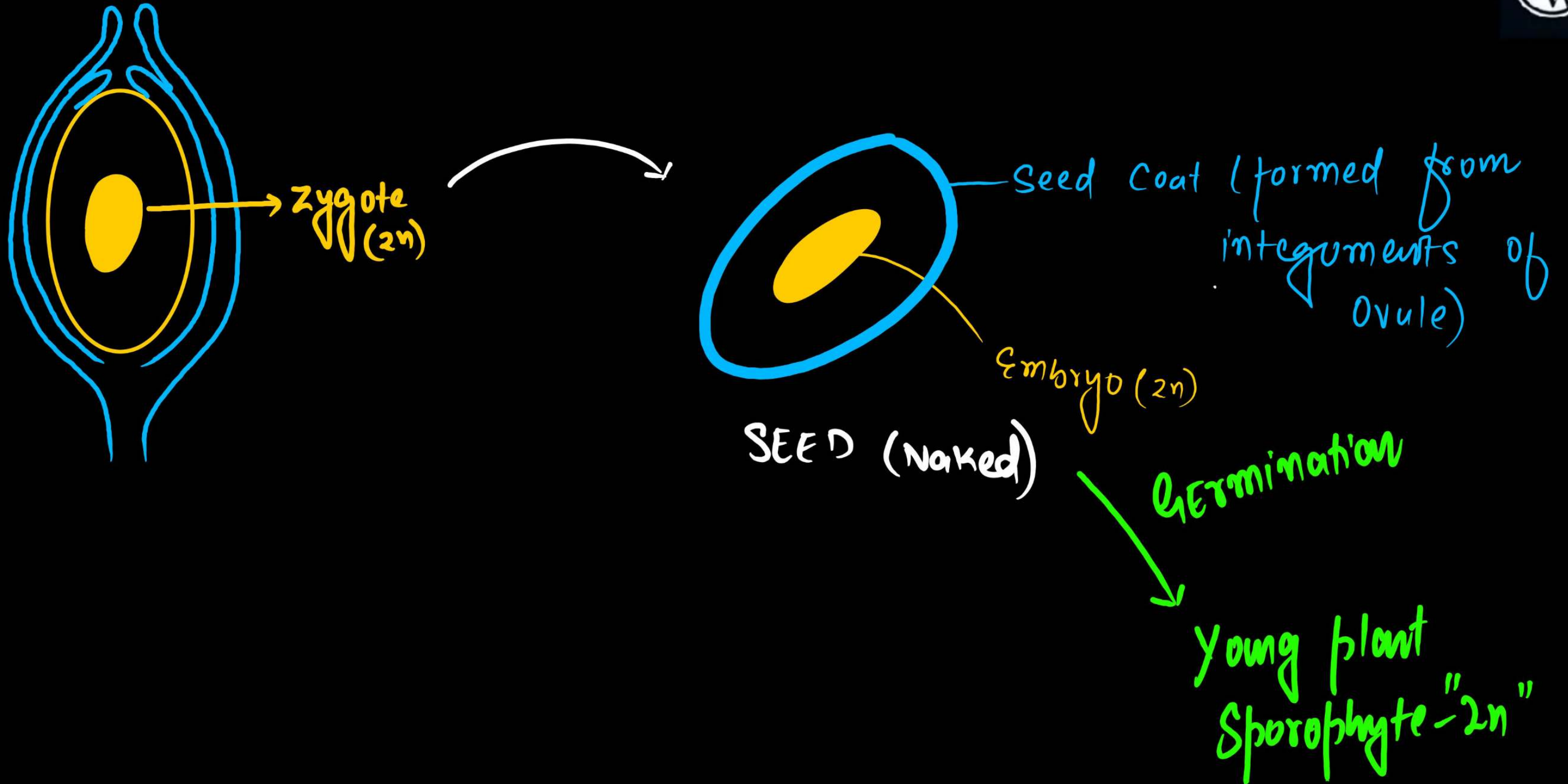
→ It occurs by wind / Air currents
kld as Anemophily

↓
Pollination by wind.



Fertilisation :-

- Pollen grain germinates and forms pollen tube which the two male gametes inside the Ovule.
- Fertilisation with the help of P.T is kld Siphonogamy.
- Normally — only one σ gamete fuses with one egg cell to form zygote ($2n$)



Polyembryony :-

presence of more than 01 embryo in a
Single seed.

→ Seen in Gymnosperms.

→ How?

↓
Ans.

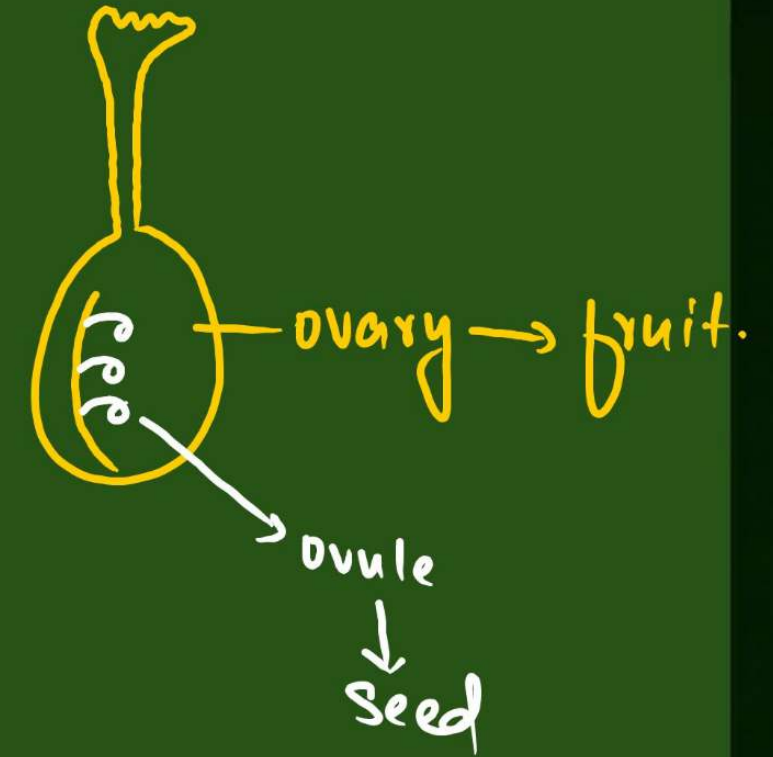
Of some times — more than one egg gets
fertilized, then polyembryony
occurs.



ANGIOSPERMS

• Angio's — Covered
sperma — seed.

• Seed is covered before and after fertilisation.



Habitat: wide range.

• Most advanced group of plant.

• Smallest Angiosperm = Wolffia.

• largest " = Eucalyptus (> 100 mtr. tall).

- Angiosperms
 - Monocots
 - Dicots.

- Angiosperms
 - Medicinal use
 - Shelter
 - Food
 - Fodder.

QUESTIONS

A gymnospermic leaf carries 16 chromosomes. The number of chromosomes in its endosperm will be: [1996]

A 12

☒ **B** 8

C 16

D 24

n

leaf $\rightarrow 2n$

$\rightarrow 2n = 16$

$2n = 16$

$n = 8$

$n = 8$

QUESTIONS

H/W

In *Pinus*, the pollen grain has 6 chromosomes then in its endosperm will have: [1992]

- A** 12
- B** 6
- C** 18
- D** 24

QUESTIONS

H/W

A plant having seeds but lacking flowers and fruits belongs to:

[1992]

- A** Pteridophytes
- B** Ferns
- C** Mosses
- D** Gymnosperms

QUESTIONS

H/w

Winged pollen grains are present in

[OS 2018]

- A** Mustard
- B** *Cycas*
- C** Mango
- D** *Pinus*

QUESTIONS

H/w

Match List-I with List-II:

Choose the correct answer from the options given below.

[MR* 2022 Re]

- A** A-II, B-III, C-I, D-IV
- B** A-III, B-I, C-II, D-IV
- C** A-III, B-IV, C-II, D-I
- D** A-III, B-II, C-I, D-IV

List-I		List-II	
A.	<i>Chlamydomonas</i>	(I)	Moss
B.	<i>Cycas</i>	(II)	Pteridophyte
C.	<i>Selaginella</i>	(III)	Alga
D.	<i>Sphagnum</i>	(IV)	Gymnosperm

QUESTIONS

H/W

Which of the following statements is incorrect about gymnosperms?

[2020-Covid]

- A** Male and female gametophytes are free living
- B** Most of them have narrow leaves with thick cuticle
- C** Their seeds are not covered
- D** They are heterosporous

QUESTIONS

H/W

In gymnosperms, the pollination is

- A** Anemophilous-micropylar
- B** Anemophilous-stigmatic
- C** Entomophilous-micropylar
- D** Entomophilous-stigmatic

QUESTIONS

Assertion (A): In gymnosperms the seeds that develop post fertilisation are naked. TRUE

Reason (R): In gymnosperms ovules are not enclosed by any ovary wall. TRUE

- ☒ **A** Both (A) and (R) are true and the (R) is the correct explanation of the (A)
- ☐ **B** Both (A) and (R) are true but the (R) is not the correct explanation of the (A)
- ☐ **C** (A) is true but (R) is false
- ☐ **D** Both (A) and (R) are false

QUESTIONS

HW

Fruits are not formed in gymnosperm because

- A** Fertilization is absent
- B** Pollination is absent
- C** Seeds are not formed
- D** Ovary is absent

HOMework

07 Questions



Thank
You